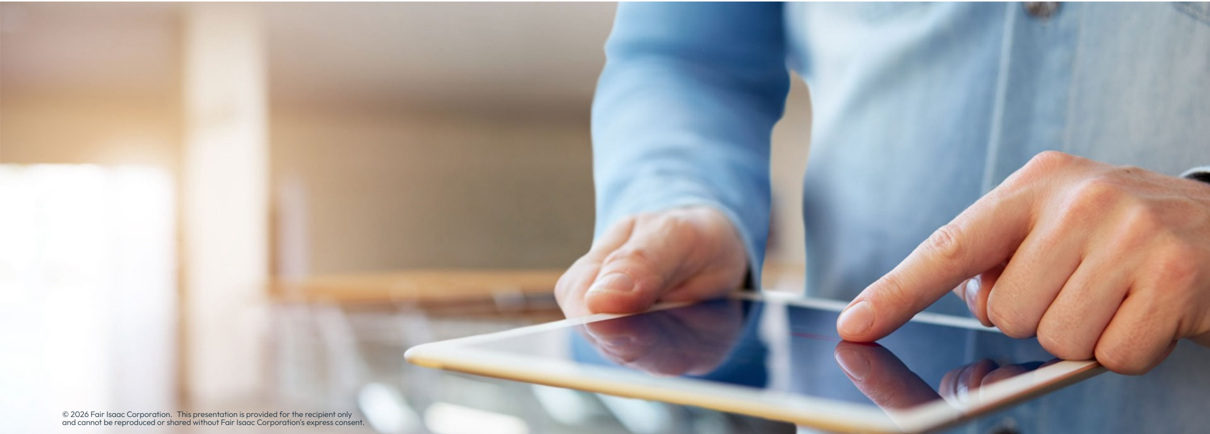
SOS constraints



Special Ordered Set (SOS) constraints

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- Special Ordered Sets (SOSs) are ordered sets of variables, where only one/two contiguous variables in the set can assume non-zero values:
  - SOS type 1 (SOS1) are a set of variables, of which at most one can take a non-zero value with all others being at zero:
    - They most frequently apply for binary variables where at most one can take the value 1
    - For example, decide the location for a new facility amongst a set of candidate locations
  - SOS type 2 (SOS2) is an ordered set of non-negative variables, of which at most two can be non-zero:
    - If two variables are non-zero, these must be consecutive in their ordering
    - Commonly used to model piecewise linear approximations of nonlinear functions

Note: *Special Ordered Sets are used by the FICO® Xpress Optimizer to improve the performance of the branch-and-bound algorithm*

Special Ordered Set (SOS) constraints

- The problem.addSOS() function can be used for creating and directly adding Special Ordered Set (SOS) constraints to a problem:

```
problem.addSOS(indices, weights, type, name)
```

  - SOS constraints enforce a small number of consecutive variables in a list to be nonzero
  - Where the arguments correspond to:
    - `indices`: list of variables composing the SOS constraint
    - `weights`: list of floating-point weights (one per variable); these define the order for SOS2 constraints, must be sufficiently distinct and and may be used in branching
    - *`type`*: type of the SOS constraint, can be 1 (default) or 2
    - *`name`*: name of the SOS constraint (optional)

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    - *`name`*: name of the SOS constraint (optional)
- Examples including Python lists for specifying `indices` and `weights`:

```
N = 20
p = xp.problem()
x = [p.addVariable() for i in range(N)]
s1 = p.addSOS([x[0], x[2]], [4,6])          # SOS type 1 with fixed weights
s2 = p.addSOS(x, [i+2 for i in range(N)], 2) # SOS type 2 with incremental weights
```

Thank You

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